

Exploratory Market Data Analysis & Alpha Signal Document

①

⇒ Data → Historical Stock price data for Apple Inc.

→ Time Series Dataset of Stock Market Data
(Used for trading strategies, Volatility Modelling & Alpha Generation)

Columns:-
Date → Trading Day
Close/Price → Final Price of the stock, that Day
Open → Price at Market Open
High/Low → Intraday Extremes
Volume → Number of Share Traded

① Daily Returns:- How much the price changes from one day to the next.

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

In Quant Finance More commonly used,

$$R_t = \ln \left(\frac{P_t}{P_{t-1}} \right)$$

② Rolling Volatility:- How much returns fluctuate i.e. the risk / uncertainty in Returns.

* Volatility is calculated over a Moving Window
e.g. ⇒ (Last 20 Days)

σ_t = Std. Deviation of Returns
Over last N days (N=20)

⇒ Short term Risk → 20 Day Rolling Volatility
⇒ Medium term Risk → 60 Day Rolling Volatility
} ⇒ High Volatility → Unstable Market
Low Volatility → Stable Market

⊛ GARCH Model is used for this Volatility Forecasting.

③ Momentum Indicator →

Whether Price trending upward or ~~would~~ downward.

'+ve' → upward trend → (Buy Signal)

'-ve' → downward trend → (Sell Signal)

$$\text{Momentum}_t = P_t - P_{t-N}$$

$$= \left(\frac{P_t}{P_{t-N}} - 1 \right)$$

One of the strongest empirical anomalies in Finance.

④ Alpha Signals → Excess Return Beyond the Market Benchmark.

Alpha Signals → Indicators that try to predict future Return.

Such As:

▷ Momentum

▷ Volatility Based Signal

▷ Mean Version

▷ Volume Shows

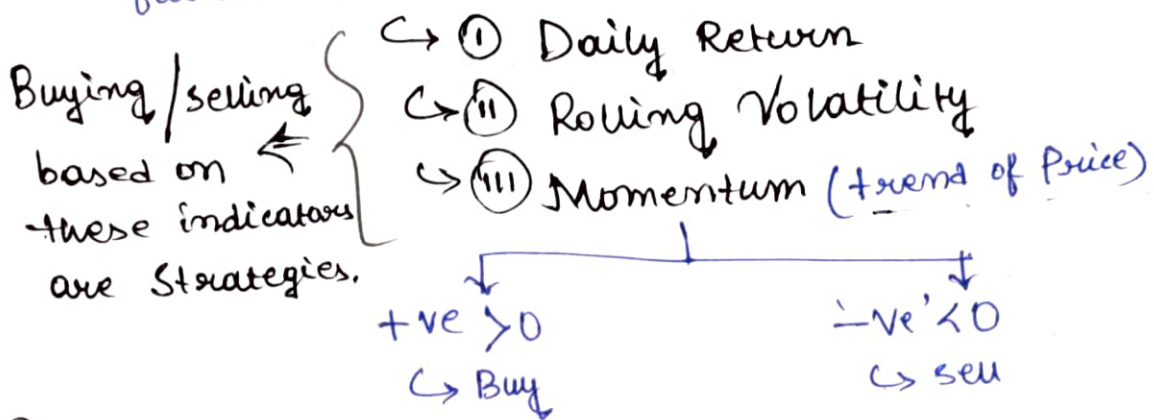
⊛ Alpha Signal is the rule that tells when to buy/sell.

Time Span of the Data
[January 2020 to December 2024]

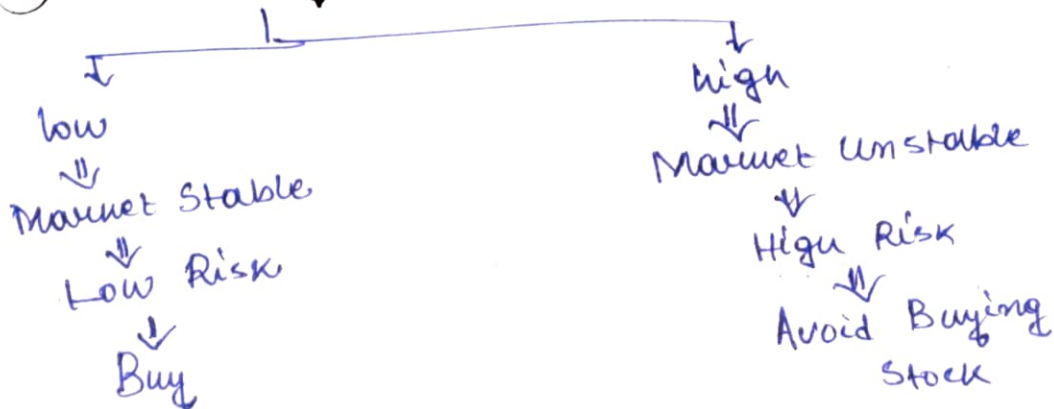
(★) How to test Simple Alpha Signals

(2)

→ Alpha Signals → Are the rules/indicators that helps us to know when to buy or sell a stock based on expected future return.



② Volatility



$$\text{CAPM: } R_t = \alpha + \beta R_{\text{market},t} + \epsilon_t$$

Extra Return not explained by the Market.

Return Explained by the Market

Signal →

+ve → Buy

0 → Stay

-ve → sell

Benchmark → Baseline Performance
↓
Return by just holding the stock for entire period.

Objective: Can past price behavior
(momentum + Volatility) helps
predicting future returns & Generate
Alpha?

Steps → ① Understand & Cleaning the Data

- (ETL)
- Load the Data
 - Check → Missing Values, Duplicate
Data type (Should not be in
String)
 - Convert Date → Date-time
→ Sort by Date

ETL → Extraction - Transform & Load

→ Process of Data Cleaning & Data struc-
turing.

Step-2 EDA → Exploratory Data Analysis
(Understanding the characteristics of Data)

→ ① Price Behavior → plot Price v/s Time
→ observe trend of prices & sudden
crashes.

→ ② Return Distribution → Compute Return
Mean = 0
↓
plot the Histogram

Fat Tail
of Histogram = Extreme Movements

→ ③ Volatility Behaviour → High v/s low
fluctuations in Cluster

Step-3 Feature Engineering → Create New Variables from the existing variables in the data which will help the Analysis. (Predict the Return)

↳ (i) Daily Return → $R_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$

↳ (ii) Rolling Volatility →

↳ (iii) Momentum indicators →

Step-4 := Statistical Pattern Analysis

(i) Auto Correlation

(ii) Volatility Clustering

(iii) Momentum Persistence

Step-5 := Build Alpha Signals

↳ Convert insights → Trading Rule

(1) Signal-1 := Momentum

$> 0 \rightarrow \text{Buy}$

else → Sell

(2) Signal-2 := Volatility

low → Buy

high → Avoid

(3) Signal-3 := Combined

↳ Momentum > 0

Volatility low.

Step-6: Backtesting → checking the reliability of my strategy / Rules.

(if I follow the Rules in Past, Would I make Money?)

① Create Signal Column $\Rightarrow$

1 = invest

0 = no invest

② Shift Signal $\rightarrow$ Signal $t-1$

③ Compute Strategy Return

$$\text{Strategy Return}_t = \text{Signal}_{t-1} \times R_t$$

④

Benchmark

Buy-and-Hold Return (Passive

(Check Look Ahead Bias) ^{Star} Strategy)

Step-7: Performance Evaluation

Metrics $\Rightarrow$ Mean Return $\Rightarrow$ Profitability

Volatility $\Rightarrow$ Risk

Sharpe Ratio $\Rightarrow$ Risk Adjusted Return

Cumulative Return

$\hookrightarrow$ Growth of investment.

Visual $\Rightarrow$ Strategy v/s Buy & Hold.

Step: 8 $\Rightarrow$ Interpret & Report Writing

★ Hands on Coding:— Python Question for interview, from Project,

① What Do you Mean by ETL ⇒

⇒ ETL → E ⇒ Extraction → Reading Raw data from Source & load in Coding Environment.

→ Transform → Data cleaning
(* imp. step)

↓
Delete/Remove Duplicate Value

II) Fill Missing values with Median

III) Data type Adjustment

↳ e.g. Convert Dates into Date Time Format.

integer i loc

We use i loc when we do integer based indexing

v/s

loc

we use loc when we do label based indexing

② Why Reset indexing

⇒ After Dropping duplicate values there are chances

of indexing Mismatch ⇒ Hence we use index reset to do a new Sequential index.
form

③ why use Date-time? ⇒ We can't keep

String values in our dataset, that's we use datetime to convert the string value date-time format, since it is required for time-based analysis.

① How to You Validate Your Data?

↳ Through data cleaning & transformation

↳ Check Missing Values.

⇒ Missing Values can distort calculations of Return, Volatility, So it's imp. to handle them before starting the Analysis.

↳ we use `isnull()` function to check Missing Values ⇒

`isnull()` ⇒ True ⇒ There Exist Missing Values in the Dataset.

↳ we use `fillna()` ⇒ to fill the Missing Values.

↳ Drop Duplicate Values ⇒ use `dropna()` ⇒ imp. as the duplicate

Values affect the integrity of the time series Analysis.

we use `duplicated()` function to check the existence of duplicate values in the dataset.

↳ It is imp. to Convert the data types before starting Analysis.

↳ we use `df.dtypes` to check the types of data stored in the columns.

⑤ Why use Log Return?

↳ Log transformation helps changing skewed distribution to Normal distribution which is imp. for Statistical Analysis & Risk Modelling.

↳ Log Returns are additive by periods

★

⑥ How you Measure/know,

How do you know that Market is Risky / unstable? $\Rightarrow$ use returns describe

$\Rightarrow$ We will check the Summary Statistics, & Analyze Statistical behavior

$\rightarrow$ if Mean of Returns $\Rightarrow$

↳ close to zero

(Market is predictable)

if Mean = Non-zero $\Rightarrow$ Market is unpredictable.

B) Std. Deviation

↳ High SD $\rightarrow$ Large Price Swing

$\downarrow$
High Volatility

↳ Low SD $\Rightarrow$ Stable Assets

↳ low Volatility

C) Min / Max, Range = (Max - Min)

High Range = Extreme Max/Min Return indicates higher Market Shock.